

Contents

In-Silico Data in Social Sciences — Use Case Analysis	1
Terminological framework	1
Use Case 1 — Silicon Sampling: LLMs as Survey Respondents	2
Use Case 2 — Automated Social Science: LLMs as Experimental Subjects	5
Use Case 3 — Multi-Agent Social Simulation (Emergent Social Dynamics)	7
Use Case 4 — Synthetic Survey Data for Privacy-Preserving Data Sharing	10
Use Case 5 — Synthetic Respondent Data for Pre-Testing Questionnaires and Study Design	13
Use Case 6 — Homo Silicus: LLMs as Economic Agents	15
Use Case 7 — Opinion Dynamics and Attitude Diffusion Modeling	17
Use Case 8 — Social System Design and Prototyping	20
Use Case 9 — Legal and Institutional Simulation	22
Use Case 10 — Synthetic Data for LLM Training (Persona-Driven Capability Generation)	24
Cross-Cutting Validity Constraints	26
Maturity Overview	27
Implications for Research Practice	28

In-Silico Data in Social Sciences — Use Case Analysis

Synthesized from Ω megaWiki (~310 papers, 129 claims, May 2026). Organized by the five-block framework: **A** Maturity · **B** Data Type · **C** Model Logic · **D** Governance · **E** Output Utilization

Terminological framework

Term	Definition (used in this document)
Reconstructed synthetic data	Derived from real datasets; preserves statistical distributions and correlations while ensuring privacy compliance
Generative synthetic data	Produced by statistical or ML models (VAEs, GANs, LLMs) without one-to-one record mapping
Structural / model-based synthetic data	Generated by rule-based or simulation methods under defined assumptions
Silicon sampling	Using LLM responses as proxies for survey respondents (Argyle et al. 2023)
Homo silicus	LLM as computational model of human economic decision-making (Horton 2023)
Algorithmic fidelity	The degree to which an LLM faithfully mirrors the response distribution of a target human population

Use Case 1 — Silicon Sampling: LLMs as Survey Respondents

Block A — Maturity

Dimension	Assessment
Problem	Survey research is expensive, slow, and structurally biased toward accessible populations (WEIRD, panel-fatigue, non-response). Rare subgroups (specific political intersections, marginalized demographics) are systematically underrepresented.
Missing data	Representative subgroup data; responses from hard-to-reach populations; longitudinal panel attrition; attitudes not yet collected in historical archives
Decision relevance	High in political science (electoral modeling, policy feedback), public health (attitude segmentation), product/UX research; lower for high-stakes decisions given validity constraints
Maturity level	Emerging — contested validity. Proof-of-concept results (2022–2023) are strong; rigorous critique (2024–2026) identifies five structural failure modes. Usable for exploration and pilot work; not ready for confirmatory inference without formal calibration.

Block B — Data Type

Dimension	Assessment
Structure	<i>Inquiry data</i> — closed-ended survey items, Likert-scale attitude measures, behavioral economics games
Granularity	Individual-level response vectors; aggregated subgroup distributions

Dimension	Assessment
Time dimension	Primarily cross-sectional; some longitudinal extrapolation (dangerous — knowledge cutoff risk)

Block C — Model Logic

Dimension	Assessment
Model type	<i>Generative synthetic data</i> via LLM with persona conditioning (demographic backstory, interview transcript, or psychographic profile)
Logic	Prompt includes identity specification → LLM generates token-level response distribution → verbalization or log-probability extraction → aggregate to population distribution
Scenarios vs. forecasts	Primarily <i>scenarios</i> (what would this persona type answer?); occasionally used for distributional forecasts

Conditioning approaches by fidelity (ascending): 1. Demographic backstory (Argyle et al. 2023) — weakest; 14–15 pp below interview 2. Psychographic profile inferred from social media (SPIRIT framework) 3. Full two-hour qualitative interview transcript (Park et al. 2024) — 85% normalized accuracy on GSS; best available

Block D — Governance

Dimension	Assessment
Data protection	Lower than real survey microdata <i>if</i> no individual can be identified from the persona specification. Risk increases with interview-grounded approaches (verbatim transcript data). GDPR relevance when using real social media content as persona seed.

Dimension	Assessment
Bias risks	Structural and well-documented: (1) Overrepresentation of liberal, educated, high-income worldviews (RLHF training); (2) belief-system over-coherence (Barrie & Cerina 2026 — 28 LLMs, robust finding); (3) heterogeneity ceiling — structural overregularization persists even after persona optimization; (4) Persona conditioning <i>degrades</i> alignment for underrepresented subgroups (Assessing Reliability 2026).
Validation available?	Yes, but method-dependent. 85% GSS accuracy under interview conditioning (GPT-4o, U.S. only); log-probability methods underestimate alignment; verbalization outperforms sampling. Cross-cultural validation largely absent (German GLES failure documented). Martingale violations (order dependence) impair Bayesian inference.

Block E — Output Utilization

Dimension	Assessment
Primary use	<i>Simulation</i> (distributional estimates for rare subgroups); <i>pilot study</i> (hypothesis generation before fielding real survey); <i>sensitivity analysis</i> (how would results change under different demographic compositions)
Appropriate scope	Exploratory, hypothesis-generating, and pilot-design phases; not confirmatory inference without statistical calibration framework (Argyle et al. 2026)
Communication	Summary statistics, synthetic crosstabs, distributional comparisons

Key papers: Argyle et al. (2023, *Political Analysis*); Horton (2023, NBER); Park et al. (2024, *Nature*); Bisbee et al. (2024); Barrie & Cerina (2026); Busby et al. (2026)

Use Case 2 — Automated Social Science: LLMs as Experimental Subjects

Block A — Maturity

Dimension	Assessment
Problem	Behavioral experiments (e.g., behavioral economics, social psychology) require human participants — costly, ethically constrained, slow, often underpowered, WEIRD-biased. Replication crisis amplifies the cost of false positives.
Missing data	Experimental evidence for rare conditions; cross-cultural replications; multi-site variation; experiments that are ethically or practically impossible with real humans
Decision relevance	Theory testing in behavioral economics; causal inference about social mechanisms; pre-registration benchmarks
Maturity level	Promising but epistemologically contested. Manning et al. (2024) demonstrate SCM-based automation; Hewitt et al. (2024, 70 pre-registered experiments) show 91% treatment-effect variation prediction. Fundamental limits: no validation against human populations in original automated studies.

Block B — Data Type

Dimension	Assessment
Structure	<i>Behavior data</i> — experimental outcomes from structured interaction protocols (bargaining, bail hearings, job interviews, auctions)
Granularity	Individual agent decision + aggregate effect-size estimates (OLS-fitted SCM coefficients)
Time dimension	Point-in-time experimental interactions; no longitudinal structure

Block C — Model Logic

Dimension	Assessment
Model type	<i>Structural / model-based synthetic data</i> — Structural Causal Models (SCMs)
Logic	operationalize hypotheses as causal graphs; factorial design generates treatment combinations; LLMs enact agent roles SCM identifies: agents (nodes), outcomes (endogenous), causes (exogenous), interaction protocol. Factorial combinations exhaust treatment space. Post-interaction survey captures LLM-reported outcomes. OLS fits SCM to data. Key insight: “The LLM knows more than it can immediately tell” — direct prompting for causal coefficients fails (13× overestimate); fitted SCM succeeds.
Scenarios vs. forecasts	<i>Scenarios</i> (if condition X, then outcome Y); causal identification possible under stated assumptions

Block D — Governance

Dimension	Assessment
Data protection	No real human data unless calibrating on real behavioral datasets. IRB requirements unclear / actively debated.
Bias risks	RLHF-trained models may produce systematically pro-social, conflict-avoidant behaviors (too polite); SCM-generated data lacks ecological validity; results may reflect training data biases rather than genuine human behavioral patterns
Validation available?	Partial: Manning et al. validate qualitatively against economic theory; no cross-species (human/LLM) quantitative comparison in original paper. Hewitt et al. provide 70-experiment validation suite.

Block E — Output Utilization

Dimension	Assessment
Primary use	<i>Simulation</i> for causal hypothesis generation; <i>pilot</i> for effect-size priors before human subject studies
Decision scope	Legitimate for theory development and pre-registration; not for policy-relevant causal inference without human validation
Communication	SCM coefficient plots; effect-size tables; counterfactual scenario comparisons

Key papers: Manning et al. (2024, automated social science); Hewitt et al. (2024, 70 experiments); Argyle et al. (2026, calibration framework)

Use Case 3 — Multi-Agent Social Simulation (Emergent Social Dynamics)

Block A — Maturity

Dimension	Assessment
Problem	Classical agent-based models (ABMs) require hand-coded behavioral rules — brittle, cannot capture nuanced linguistic/reasoning behaviors, require expert calibration. Social dynamics (polarization, norm diffusion, policy shock response) are hard to study ethically in the real world.
Missing data	High-resolution behavioral data on social interaction under counterfactual conditions (policy changes, external shocks, novel technologies)
Decision relevance	Policy simulation (UBI effects, health interventions); platform design (content moderation impacts); crisis response planning (pandemic, disaster)

Dimension	Assessment
Maturity level	Emerging infrastructure. Park et al. (2023, Generative Agents) established the core memory-reflection-planning architecture. AgentSociety (2025) demonstrated 10,000-agent scale. Concordia (Google, 2023) provides general-purpose framework. Qualitative pattern validation is available; quantitative validation is rare and limited.

Block B — Data Type

Dimension	Assessment
Structure	<i>Behavior data</i> — agent actions (mobility, communication, economic exchange, social interaction); <i>structural data</i> — network topology, institutional rules
Granularity	Individual agent action + meso-level emergent patterns (polarization index, mobility statistics, health outcomes) + macro-level trend alignment
Time dimension	Dynamic — multi-step simulations; emergent temporal patterns (information diffusion curves, polarization growth, shock recovery)

Block C — Model Logic

Dimension	Assessment
Model type	<i>Structural / model-based synthetic data</i> — agents are LLM-backed; environment encodes institutional rules, network structure, physical constraints

Dimension	Assessment
Architecture	Memory-Reflection-Planning (Park et al.): memory stream (natural language, recency-importance-relevance retrieval) → periodic reflection → hierarchical daily/hourly planning → reactive action. AgentSociety adds: dual-stream memory (Event Flow + OCC-model appraisals), Maslow hierarchy for need-driven behavior, Ray distributed computing (43K MQTT msg/s). Concordia : Game Master architecture separates agent cognition from environment physics; component-based working memory.
Scenarios vs. forecasts	<i>Scenarios</i> — counterfactual policy conditions, exogenous shocks. Not forecasts in the predictive sense (no quantitative calibration to real-time data).

Block D — Governance

Dimension	Assessment
Data protection	Agents are synthetic; risk arises if initialization uses real individual-level microdata for calibration (requires pseudonymization)
Bias risks	RLHF training produces overly pro-social, conflict-avoidant agents; instruction-tuning suppresses extremist/fringe behaviors; English/US-centric behavioral priors; economic simplifications (no goods/labor markets in AgentSociety); polarization dynamics may be systematically muted
Validation available?	Qualitative pattern alignment (AgentSociety: UBI effects align with Texas trial, hurricane mobility aligns with real data). Quantitative calibration absent. Concordia acknowledges no validation in paper.

Block E — Output Utilization

Dimension	Assessment
Primary use	<i>Simulation</i> — policy scenario analysis; emergence mapping; stress-testing institutional designs
Appropriate scope	Qualitative insights and hypotheses; not quantitative prediction. Strong for: (a) identifying mechanisms not predictable from agent-level rules; (b) stress-testing counterfactual policies; (c) generating hypotheses for real-world study
Communication	Time-series plots of emergent measures; agent trajectory visualizations; counterfactual comparison tables

Key papers: Park et al. (2023, Generative Agents, UIST); Park et al. (2022, Social Simulacra); Zhao et al. (2023, Concordia); Pang et al. (2025, AgentSociety); Tessler et al. (2025, LLM attitude diffusion)

Use Case 4 — Synthetic Survey Data for Privacy-Preserving Data Sharing

Block A — Maturity

Dimension	Assessment
Problem	Social survey microdata (census, panel studies, administrative registers) cannot be shared openly due to re-identification risk. Statistical disclosure control (suppression, rounding, perturbation) degrades analytical utility.
Missing data	Shareable microdata at individual record level; rare-event data with small cell sizes; cross-national comparative data restricted under data protection regimes
Decision relevance	High for public statistics agencies, academic data archives, cross-national research; compliance with GDPR, US Privacy Act, HIPAA analogs

Dimension	Assessment
Maturity level	Established for reconstructed synthetic data (classical differential privacy, CTGAN, copula-based synthesis). LLM-based generation for this purpose is at early research stage — promising but analytically unvalidated for complex survey microstructures.

Block B — Data Type

Dimension	Assessment
Structure	<i>Inquiry data</i> — survey microdata (demographic attributes, attitudes, behaviors, income); <i>administrative data</i> — registry linkages
Granularity	Record-level individual units; preserving marginal distributions, joint distributions, and analytic utility (regression coefficients, subgroup estimates)
Time dimension	Primarily cross-sectional release; repeated cross-sections (e.g., annual SOEP/GLES releases); panel data synthesis remains harder

Block C — Model Logic

Dimension	Assessment
Model type	<i>Reconstructed synthetic data</i> (classical approach: differential privacy, CTGAN, sequential synthesis, copulas) or <i>generative synthetic data</i> (LLM fine-tuned on microdata — Polypersona approach; German General Personas from ALLBUS)

Dimension	Assessment
Logic	Classical: fit distributional model to real data → sample synthetic records → validate utility (U-score: regression coefficient similarity, propensity overlap). LLM-based: fine-tune with LoRA on microdata records → generate personas → validate against held-out real respondents
Scenarios vs. forecasts	<i>Data substitutes</i> — no scenario projection; goal is empirical fidelity, not counterfactual generation

Block D — Governance

Dimension	Assessment
Data protection	Core objective. Differential privacy provides formal guarantee (ϵ -DP). LLM-based synthesis (Polypersona) uses LoRA + 4-bit quantization; Synonymix provides privacy-by-design via knowledge graph merging (max 13% source contribution per merged node).
Bias risks	Classical synthesis: rare subgroups compressed or suppressed. LLM synthesis: training data biases imported; Polypersona evaluated only against LLM-generated references — circular validation risk.
Validation available?	Classical: established utility metrics (U-score, R-U map, propensity overlap). LLM-based: BERTScore high (>0.88) but validated against synthetic references, not real respondents — methodologically weak . German General Personas validated against ALLBUS classifiers.

Block E — Output Utilization

Dimension	Assessment
Primary use	<i>Data sharing / dissemination</i> — open research datasets, cross-institutional collaboration, teaching datasets
Appropriate scope	Secondary analysis, methodological research, teaching; not primary empirical inference about real populations without independent validation of synthetic vs. real data equivalence
Communication	Released microdata files; data quality reports; utility-privacy trade-off documentation

Key papers: Busby et al. (2025, more parameters than populations); Colglazier et al. (2025, Polypersona); Huber et al. (2025, German General Personas); Brinkmann & Yáñez (2026, Synonymix)

Use Case 5 — Synthetic Respondent Data for Pre-Testing Questionnaires and Study Design

Block A — Maturity

Dimension	Assessment
Problem	Pre-testing survey instruments (cognitive interviews, pilot surveys) is expensive and slow. Researchers need to evaluate question wording, order effects, response option design, and likely distributions before fielding costly real studies.
Missing data	Pre-field estimates of likely response distributions; cognitive burden indicators; wording effects
Decision relevance	Survey methodology; questionnaire design optimization; sampling frame decisions
Maturity level	Early-to-developing. LLMs are used informally by practitioners for question wording. Formal evaluation of this use case is limited; “Survey Response Generation” paper (Veselovsky et al. 2025, 32M responses × 8 SRGMs) provides most systematic evidence.

Block B — Data Type

Dimension	Assessment
Structure	<i>Inquiry data</i> — draft survey items, response option sets
Granularity	Item-level response distribution estimates; potential mode effects; flagging likely ceiling/floor effects
Time dimension	Point-in-time (pre-fielding stage)

Block C — Model Logic

Dimension	Assessment
Model type	<i>Generative synthetic data</i> via LLM with or without persona conditioning
Key finding	Survey Response Generation Method (SRGM) choice is the dominant determinant of apparent alignment — not persona specification or model size. Restricted generation outperforms sampling; chain-of-thought reasoning inconsistently helps. Prior work silently varied SRGMs, making cross-study comparisons unreliable.
Scenarios vs. forecasts	Wording effect scenarios; distributional forecasts (weak evidence base)

Block D — Governance

Dimension	Assessment
Data protection	Low risk — no real individual data; LLM responses are synthetic
Bias risks	SRGM choice can change substantive conclusions about which subgroups are well-simulated; liberal/educated priors inflate apparent alignment for some subgroups
Validation available?	Yes (Veselovsky et al. 2025 is definitive here): 10 LLMs × 8 SRGMs × 4 real surveys, demonstrating SRGM sensitivity

Block E — Output Utilization

Dimension	Assessment
Primary use	<i>Decision support</i> for questionnaire revision; <i>simulation</i> of likely distributions for power calculations; <i>communication</i> of expected response variation to study teams
Appropriate scope	Questionnaire development and power analysis; not a substitute for actual cognitive interviewing or pilot fielding

Key papers: Veselovsky et al. (2025, Survey Response Generation); Tjua et al. (2024, distributional alignment benchmarking)

Use Case 6 — Homo Silicus: LLMs as Economic Agents

Block A — Maturity

Dimension	Assessment
Problem	Behavioral economics experiments require human participant pools for each treatment; calibrating heterogeneous agent types across many game configurations is intractable at scale. Economic theorists lack computational tools for rapid theory testing.
Missing data	Behavioral responses under rare parametric conditions; cross-game generalization of calibrated types
Decision relevance	Economic theory building; policy impact modeling (minimum wage, welfare reforms); institutional design
Maturity level	Promising — reproducible in narrow scope. Horton (2023) establishes methodology; qualitative replications of behavioral economics classics confirmed. Calibration-and-generalization approach shows strong within-paradigm results. No quantitative cross-human validation in main paper.

Block B — Data Type

Dimension	Assessment
Structure	<i>Behavior data</i> — experimental game outcomes (dictator allocations, minimum wage choices, status quo bias magnitudes, prospect theory parameters)
Granularity	Individual agent decision (token-level distribution) → calibrated mixture-of-types distribution → aggregate effect
Time dimension	Point-in-time strategic interactions; no intertemporal dynamics in canonical experiments

Block C — Model Logic

Dimension	Assessment
Model type	<i>Structural / model-based synthetic data</i> — economic game structure defines the interaction; LLM response functions as agent decision logic
Calibration approach	Mixture-of-types model fitted to in-distribution game data; tested on structurally different games (out-of-distribution). Calibrated agents reduce MSE from 0.182 → 0.094 vs. persona-less baselines.
Scenarios vs. forecasts	Primarily <i>scenarios</i> — parametric variation within game structure; limited cross-game forecast generalization

Block D — Governance

Dimension	Assessment
Data protection	No real individual data; LLM responses are synthetic

Dimension	Assessment
Bias risks	RLHF training suppresses genuine conflict-seeking, exploitation-seeking, and purely self-interested behavior; pro-social bias documented. Memorization risk: LLMs may produce game-theoretically “correct” results from training data rather than behavioral responses.
Validation available?	Qualitative: 5/5 behavioral economics classics replicated (dictator games, fairness, status quo bias, prospect theory, min wage). Quantitative calibration MSE reduction documented. No comparison to actual human participants across same games.

Block E — Output Utilization

Dimension	Assessment
Primary use	<i>Simulation</i> of behavioral economic mechanisms; <i>hypothesis generation</i> for experimental designs; <i>sensitivity analysis</i> on parametric variations impractical at human-subject scale
Appropriate scope	Hypothesis generation and theory refinement; not causal inference about real economic behavior
Communication	Game outcome distributions; calibrated type profiles; effect magnitude comparisons to behavioral economics literature

Key papers: Horton (2023, Homo silicus, NBER); Manning et al. (2024, automated social science)

Use Case 7 — Opinion Dynamics and Attitude Diffusion Modeling

Block A — Maturity

Dimension	Assessment
Problem	How attitudes spread, polarize, or converge through social networks is difficult to study empirically: it requires repeated individual-level attitude measurement during natural social interaction, which is impractical at scale. Classical opinion dynamics models (Deffuant, Hegselmann-Krause) use toy behavioral rules.
Missing data	Mechanism-level data on how individual interactions translate to opinion change; counterfactual network topologies
Decision relevance	Platform content policy (filtering, recommendation algorithms); counter-disinformation strategy; political communication
Maturity level	Early research. Tessler et al. (2026) demonstrate LLM-backed agents in small-world networks with live news retrieval; AgentSociety shows polarization dynamics at scale. No quantitative ground-truth validation of diffusion dynamics.

Block B — Data Type

Dimension	Assessment
Structure	<i>Behavior data</i> — agent opinion states over time; <i>structural data</i> — network topology; <i>inquiry data</i> — attitude item trajectories
Granularity	Individual opinion vector (multidimensional attitudes) + network-level polarization indices + population distribution shifts
Time dimension	Dynamic — temporal opinion trajectories; convergence/divergence curves

Block C — Model Logic

Dimension	Assessment
Model type	<i>Structural / model-based synthetic data</i> — network topology defines interaction probabilities; LLM agents execute
Logic	Observe-Think-Act loops with belief update Real-time news retrieval augments agent context → agent interprets news through existing belief system → interaction with neighbor agents → belief update based on discourse → repeat. Homophilic vs. heterophilic network topology as key experimental parameter.
Scenarios vs. forecasts	<i>Scenarios</i> — what happens to polarization under different network topologies or information diets? Not calibrated forecasting.

Block D — Governance

Dimension	Assessment
Data protection	Agents are synthetic; risk if network initialization uses real user-graph data
Bias risks	RLHF models are systematically too moderate; extremist/fringe opinion dynamics are systematically underrepresented; heterogeneous belief representations impaired (Barrie & Cerina 2026)
Validation available?	AgentSociety: homophilic network 52% more polarized vs. heterophilic — directionally consistent with empirical literature but not quantitatively calibrated. No comparison to real online deliberation dynamics.

Block E — Output Utilization

Dimension	Assessment
Primary use	<i>Simulation</i> for platform design decisions; <i>scenario analysis</i> for regulatory impact assessment

Dimension	Assessment
Appropriate scope	Mechanism illustration and hypothesis generation; not quantitative prediction of real-world opinion shifts
Communication	Opinion trajectory plots; polarization index time-series; network topology sensitivity analyses

Key papers: Tessler et al. (2026, attitude diffusion); Pang et al. (2025, AgentSociety); Park et al. (2023, information diffusion in Generative Agents)

Use Case 8 — Social System Design and Prototyping

Block A — Maturity

Dimension	Assessment
Problem	Designing online communities, content moderation systems, or participatory platforms requires understanding how diverse populations will interact with and within them — before the platform exists. Standard user testing is expensive and cannot cover rare or adversarial user behaviors.
Missing data	Pre-deployment interaction data; adversarial behavior patterns; norm-violation responses; community formation dynamics
Decision relevance	Platform design; content policy; HCI research; civic technology
Maturity level	Operational. Social Simulacra (Park et al. 2022) is the canonical application: Generate (populate community) + WhatIf (counterfactual what-if scenarios). Designers use it iteratively; Turing-test studies show synthetic behaviors are often indistinguishable from real. Well-suited because it doesn't require distributional accuracy — design iteration only needs relative/qualitative insight.

Block B — Data Type

Dimension	Assessment
Structure	<i>Behavior data</i> — posts, replies, anti-social behaviors, norm violations, community formation
Granularity	Individual interaction episodes; community-level interaction patterns; policy counterfactuals
Time dimension	Session-level simulation; no longitudinal continuity requirement

Block C — Model Logic

Dimension	Assessment
Model type	<i>Structural / model-based synthetic data</i> — community rules define the interaction frame; LLM agents populate it
Logic	Designer specifies community purpose and rules → LLM generates diverse synthetic members → simulates interaction → Generate shows candidate behaviors → WhatIf replaces an actor or rule to branch the scenario
Scenarios vs. forecasts	<i>Counterfactual scenarios</i> — what happens if we change this rule? Not population-representative forecasts.

Block D — Governance

Dimension	Assessment
Data protection	Low risk — no real user data required
Bias risks	LLMs may fail to simulate genuinely extremist users; fringe / minority behaviors underrepresented; cultural contexts outside training data poorly handled
Validation available?	Turing-test with Prolific users: synthetic behaviors often indistinguishable from real. But validity criterion is <i>design plausibility</i> , not distributional accuracy — a less demanding standard

Block E — Output Utilization

Dimension	Assessment
Primary use	<i>Simulation</i> for iterative design; <i>decision support</i> for platform policy choices; <i>communication</i> of design scenarios to stakeholders
Appropriate scope	Design iteration and discovery; not prediction of actual user behavior at scale

Key papers: Park et al. (2022, Social Simulacra, UIST)

Use Case 9 — Legal and Institutional Simulation

Block A — Maturity

Dimension	Assessment
Problem	Legal system dynamics (crime, prosecution, sentencing, corruption) evolve over long time horizons and are hard to study experimentally. Policy changes (sentencing reform, police deployment) have diffuse, lagged, and interactive effects that ABM can model but classical ABMs have lacked behavioral realism.
Missing data	High-resolution behavioral data on individual legal decision-making; counterfactual crime/prosecution scenarios under different institutional rules
Decision relevance	Criminal justice policy; legal system reform; institutional design
Maturity level	Early research. Law in Silico (2025) is the primary demonstration: hierarchical LLM framework, real crime statistics for initialization. Qualitative validation only.

Block B — Data Type

Dimension	Assessment
Structure	<i>Behavior data</i> — individual crime/compliance decisions; <i>structural data</i> — institutional rules, sentencing guidelines, network of legal actors
Granularity	Individual agent decisions + aggregate crime trends + institutional corruption indices
Time dimension	Dynamic — multi-period simulation of policy interventions

Block C — Model Logic

Dimension	Assessment
Model type	<i>Structural / model-based synthetic data</i> — hierarchical LLM agent framework; real statistical distributions used for initialization (grounded in actual crime data)
Logic	Agents represent citizens, police, prosecutors, judges with role-specific decision functions; institutional rules encoded as game-master constraints; emergent dynamics from agent interactions
Scenarios vs. forecasts	<i>Scenarios</i> — policy counterfactuals (sentencing reform, police deployment reallocation)

Block D — Governance

Dimension	Assessment
Data protection	Aggregate crime statistics used for initialization — no individual-level data required
Bias risks	LLMs reflect training data biases about criminality, policing, and legal systems; racial/class biases documented in NLP applications likely persist in legal simulation; RLHF suppresses genuinely adversarial or corrupt behavior

Dimension	Assessment
Validation available?	Qualitative: macro-level crime trends reproduced; micro-level cat-and-mouse behaviors emerge; institutional corruption patterns align directionally. No quantitative comparison to real criminal justice statistics.

Block E — Output Utilization

Dimension	Assessment
Primary use	<i>Simulation</i> for policy scenario analysis; <i>communication</i> of mechanisms to policymakers
Appropriate scope	Exploratory policy analysis; not quantitative prediction

Key papers: Cai et al. (2025, Law in Silico)

Use Case 10 — Synthetic Data for LLM Training (Persona-Driven Capability Generation)

Block A — Maturity

Dimension	Assessment
Problem	Training data diversity is a bottleneck for LLM capability: models trained on uniform or narrow data generalize poorly. Human-generated diverse data is expensive at scale.
Missing data	Diverse training examples covering rare domains, cultural contexts, demographic perspectives
Decision relevance	LLM training pipelines; benchmark construction; instruction-following alignment

Dimension	Assessment
Maturity level	Operational. PersonaHub (Ge et al. 2024) demonstrates at scale: 1B synthetic personas auto-curated from web data → persona-driven data synthesis → 7B model trained on persona-math achieves 65% MATH, matching GPT-4-Turbo at the time. 350 citations in one year.

Block B — Data Type

Dimension	Assessment
Structure	<i>Inquiry / behavior data</i> — synthetic Q&A pairs, mathematical reasoning chains, diverse instruction-following examples
Granularity	Instance-level training examples; population-diverse coverage
Time dimension	Static corpus (no temporal structure)

Block C — Model Logic

Dimension	Assessment
Model type	<i>Generative synthetic data</i> — billion-scale persona collection → persona-conditioned synthesis of diverse examples
Logic	Large-scale web text → profile extraction → 1B persona embeddings → persona-conditioned data generation → downstream model fine-tuning. Diversity is the key property, not alignment to a real population distribution.
Scenarios vs. forecasts	Not applicable — purpose is capability, not prediction

Block D — Governance

Dimension	Assessment
Data protection	Web-scraped persona data may contain personal information; extraction pipeline requires deidentification
Bias risks	Amplifies whatever biases are present in web training corpus; capability gains may embed skewed worldviews at scale; enables rapid capability distillation (ethical concern flagged in wiki)
Validation available?	Strong benchmark validation: MATH, MMLU, instruction following benchmarks show consistent gains over non-diverse synthesis

Block E — Output Utilization

Dimension	Assessment
Primary use	<i>LLM training data</i> — not a social science analysis tool per se; but capabilities built on persona-diverse data directly affect quality of all use cases 1–9 above
Appropriate scope	Capability training and benchmarking

Key papers: Ge et al. (2024, PersonaHub/Scaling Synthetic Data); Colglazier et al. (2025, Polypersona)

Cross-Cutting Validity Constraints

The following structural failure modes apply across most use cases and set hard limits on what in-silico social data can deliver:

Failure Mode	Mechanism	Affected Use Cases
Belief-system over-coherence	LLMs produce unrealistically high inter-attitude correlations; ideology is over-concentrated on first principal component (Barrie & Cerina 2026, 28 LLMs)	1, 2, 6, 7

Failure Mode	Mechanism	Affected Use Cases
Heterogeneity ceiling	Structural overregularization persists even after persona optimization; no segmentation strategy fully restores within-group variance	1, 3, 7
Martingale violations	Response distributions depend on question order; violates exchangeability assumption required for Bayesian inference	1, 5
SRGM sensitivity	Survey Response Generation Method choice dominates alignment; prior comparisons unreliable due to uncontrolled method variation	1, 5
Cross-cultural failure	Most validation is US/English/WEIRD; documented failures for German political opinion (GLES); conditioning degrades further for underrepresented subgroups	1, 2, 6
RLHF pro-social bias	Fine-tuning suppresses extremist, conflict-seeking, and selfish behaviors that are sociologically consequential	3, 7, 9
Training data contamination	LLMs may produce “correct” outcomes by retrieving training examples rather than generalizing behavioral rules	2, 6

Maturity Overview

Use Case	Maturity	Ready for confirmatory use?
1 — Silicon Sampling / Survey Simulation	Emerging — contested	No (only with formal statistical calibration)
2 — Automated Experimental Social Science	Promising	No (exploration/pilot only)
3 — Multi-Agent Social Simulation	Emerging infrastructure	No (qualitative insights only)
4 — Privacy-Preserving Data Synthesis	Established (classical); early (LLM-based)	Classical: Yes. LLM-based: No
5 — Questionnaire Pre-Testing	Early-to-developing	Partial (SRGM-controlled design only)
6 — Homo Silicus / Economic Agents	Promising — narrow scope	No (hypothesis generation only)
7 — Opinion Dynamics Modeling	Early research	No (mechanism illustration only)
8 — Social System Design Prototyping	Operational	Yes (design iteration, not population prediction)
9 — Legal/Institutional Simulation	Early research	No (exploratory only)
10 — Synthetic LLM Training Data	Operational	Yes (capability benchmarking)

Implications for Research Practice

What in-silico data can already do well: - Rapid questionnaire iteration and pilot design (Use Cases 5, 8) - Hypothesis generation before expensive human-subject studies (UC 2, 6) - Exploring mechanism dynamics under counterfactual conditions (UC 3, 7, 9) - Populating and stress-testing platform designs (UC 8) - Generating training data diversity for LLM capability (UC 10) - Recreating known subgroup distributional patterns with interview-grounded conditioning (UC 1 under specific conditions)

What requires additional development: - Formal statistical calibration frameworks to enable confirmatory inference (UC 1, 2) - Cross-cultural and non-WEIRD validation corpora - Methods to restore heterogeneity at the belief-system structure level (not just marginals) - Quantitative calibration of agent-based simulations against real temporal panel data (UC 3, 7) - Independent validation of LLM-based synthetic data generation against real microdata (UC 4)

Governance priorities: - Mandatory transparency reporting on persona specification (Whose Personae? checklist: Application, Population, Data Source, Ecological Validity, Reproducibility, Generalizability) - SRGM pre-registration to enable cross-study comparisons - Subgroup

fairness reporting (Demographic Parity Difference) for all simulation studies - Clear disclosure distinguishing synthetic data from human-collected data in all publications

Sources: OmegaWiki in-silico social science corpus, ~310 papers, synthesized 2026-05-29. Key foundational papers: Argyle et al. 2023 (silicon sampling); Horton 2023 (homo silicus); Park et al. 2023 (generative agents); Park et al. 2024 (1,000 people simulation); Manning et al. 2024 (automated social science); Barrie & Cerina 2026 (belief system distortion); Bisbee et al. 2024 (martingale violations); Veselovsky et al. 2025 (SRGM sensitivity).